GLOBAL FOOD SECURITY & AGRICULTURAL COMMODITY ANALYSIS DASHBOARD

Project Overview

The Global Food Security & Agricultural Commodity Analysis Dashboard is an interactive Business Intelligence solution developed using **Python (JupyterLite)** for data cleaning and preprocessing and **Power BI** for data visualization and reporting.
The objective of this project is to analyze the relationship between agricultural commodity prices, crop production, fertilizer costs, and food security indicators. The dashboard helps stakeholders understand market trends, agricultural performance, and food security risks through dynamic visualizations and KPI metrics.

Business Problem Statement

Food security is influenced by several interconnected factors including:
Commodity price fluctuations
Fertilizer cost variations
Crop production levels
Agricultural productivity
Food affordability
Crisis events and supply stress
Organizations and policymakers require a centralized dashboard to monitor these indicators and identify trends that may impact food security and agricultural sustainability.

Project Objectives

The primary objectives of this project are:
Analyze agricultural commodity price trends.
Monitor crop production and agricultural performance.
Evaluate fertilizer cost impacts.
Track food security indicators.
Identify food crisis and supply stress periods.
Generate actionable insights through visual analytics.
Support data-driven decision making.

Tools & Technologies Used

Tool	Purpose
JupyterLite	Data Cleaning & Feature Engineering
Python	Data Processing
NumPy	Numerical Computations
Pandas	Data Manipulation
Power BI Desktop	Dashboard Development
DAX	KPI Measure Creation

Dataset Description

The project utilizes four datasets:

1. Commodity Prices Dataset

Records

780 Rows × 18 Columns

2. Crop Yields Dataset

Records

64 Rows × 11 Columns

3. Food Security Indicators Dataset

Records

63 Rows × 8 Columns

4. Food Security Panel Dataset

Records

420 Rows × 39 Columns

Data Cleaning & Preprocessing

Data preprocessing was performed using **JupyterLite**.

Steps Performed

1. Imported Required Libraries

import pandas as pd

import numpy as np

2. Loaded CSV Files

Datasets were imported into Pandas DataFrames.

3. Data Inspection

Performed:

df.head()

df.info()

df.describe()

df.shape

4. Missing Value Treatment

Identified missing values

Replaced missing values using appropriate methods

5. Date Formatting

Converted date columns into datetime format.

6. Feature Engineering

Created additional indicators:

MOM Percentage Change

YOY Percentage Change

Fertilizer Composite Index

Food Shock Score

Crisis Flags

Supply Stress Flags

DAP-to-Wheat Ratio

7. Data Export

Cleaned datasets were exported as CSV files and imported into Power BI.

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DAX Measures Created

The following KPI measures were created:

Average Food Price Index = AVERAGE(food_security_panel_cleaned[ffpi_overall])

Average Wheat Price =AVERAGE(commodity_prices_cleaned[wheat_usd_mt])

Average Maize Price = AVERAGE(commodity_prices_cleaned[maize_usd_mt])

Average Fertilizer Index = AVERAGE(food_security_panel_cleaned[fertiliser_composite_idx])

Average Crop Yield = AVERAGE(food_security_indicators_cleaned[cereal_yield_kg_ha])

Average MOM Growth % = AVERAGE(food_security_panel_cleaned[ffpi_overall_mom_pct])

Average YOY Growth % = AVERAGE(food_security_panel_cleaned[ffpi_cereals_yoy_pct])

Crisis Count = SUM(food_security_panel_cleaned[ffpi_crisis_flag])

Supply Stress Count = SUM(food_security_panel_cleaned[supply_stress_flag])

Dashboard Pages

Page 1: Executive Summary

KPIs

Average Food Price Index
Average Wheat Price
Average Maize Price
Average Fertilizer Index
Average Crop Yield
MOM Growth %
YOY Growth %
Crisis Count

Visuals

Food Price Index Trend
Food Shock Score Trend

Page 2: Commodity Analysis

KPIs

Average Wheat Price
Average Maize Price
Average Fertilizer Index
Highest Commodity

Visuals

Commodity Price Trend
Fertilizer Composite Index Trend
DAP-to-Wheat Ratio Trend
Food Price YOY Growth Trend

Page 3: Agriculture Analysis

KPIs

Average Crop Yield
Highest Production
Lowest Production
Average Cereal Yield

Visuals

Crop Production Trend
Total Staple Production
Fertilizer Index Trend

Crop Area Trend

Page 4: Food Security Analysis

KPIs

Crisis Count

Supply Stress Count

Average Food Shock Score

Average Undernourishment %

Visuals

Food Crisis Events by Year

Supply Stress Events by Year

Undernourishment Trend

Food Production Index Trend

Key Insights

Commodity Analysis

Soybean Oil recorded the highest average market price.

Fertilizer prices showed significant volatility after 2008.

Commodity prices experienced major fluctuations during global economic disruptions.

Agriculture Analysis

Wheat contributed the highest production among staple crops.

Total staple production increased steadily over time.

Agricultural land allocation remained relatively stable.

Food Security Analysis

Food crisis events increased during periods of price shocks.

Supply stress events aligned with fertilizer price increases.

Food production index demonstrated long-term growth.

Executive Summary

Average Food Price Index remained relatively stable.

Crisis and supply stress events impacted affordability.

Agricultural productivity improved despite commodity volatility.

Conclusion

This project successfully integrated commodity prices, agricultural production data, fertilizer indicators, and food security metrics into a unified analytical dashboard. Using Python for preprocessing and Power BI for visualization enabled comprehensive analysis of global food security trends and agricultural market dynamics.

The dashboard provides stakeholders with actionable insights to monitor food security conditions, evaluate agricultural performance, and support informed decision-making for sustainable food systems.